

Access control and monitoring

DRAFT INSTRUCTION - WORK IN PROGRESS

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Access Control

Specification

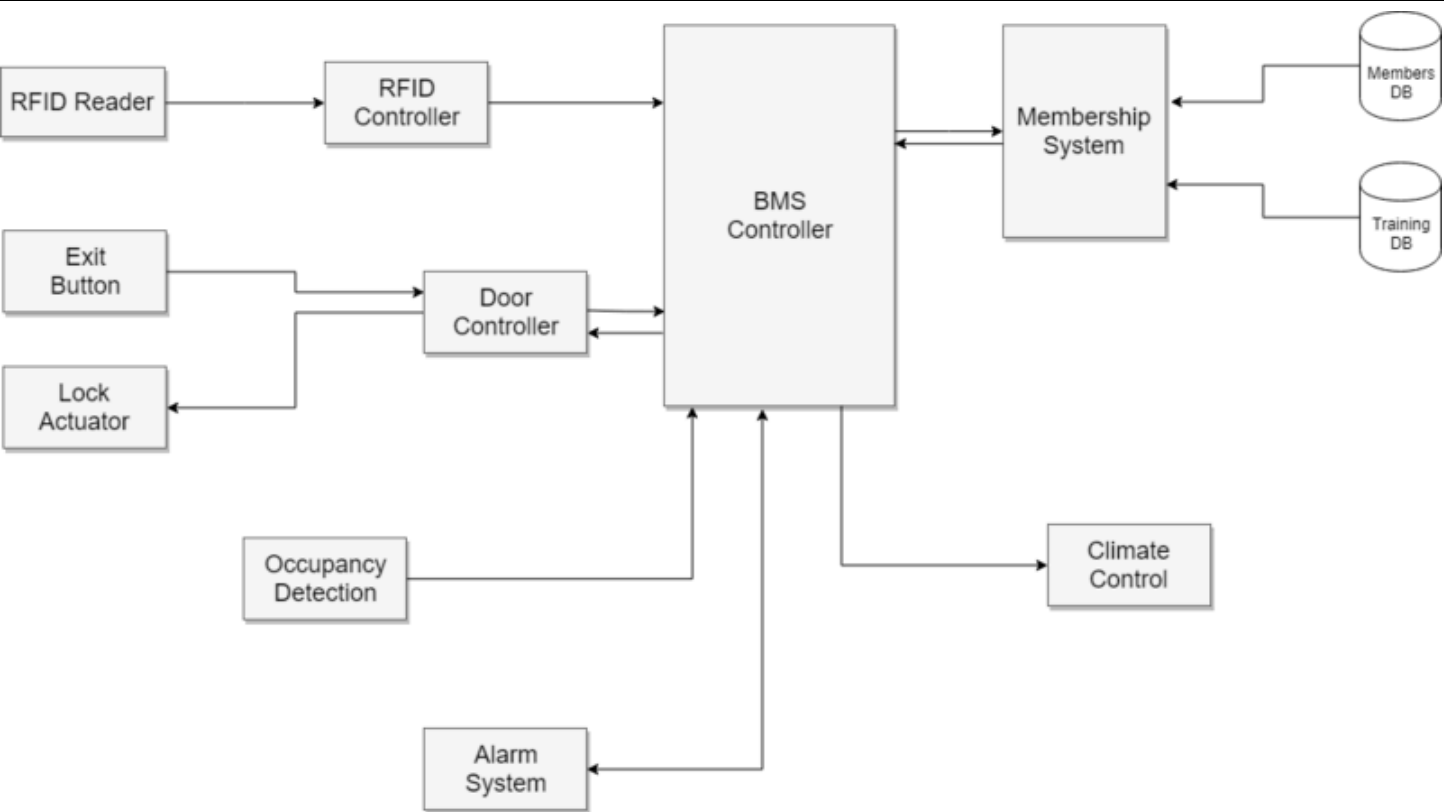
Spec needs defining.

Description

- We want some sort of system to capture an RFID identifier, store it, associate it with a person (in such a way that we can later tie it into some sort of presently-non-existent membership database).
- It needs to interface with the door, which has some sort of radio remote control, and I guess we also need some lock/unlock buttons in the space itself to stop people manually driving the door lock (as it then becomes out of sync with itself).
- With a magnetic close sensor so that when the door is in the closed position it always relocks, but with an override (for Monday nights, for example)
- A very detailed spec for Lancaster University is available for inspiration here:
<https://www.lancaster.ac.uk/iss/info/supplierdocs/network/LU-Access-Control-Spec-v1c.pdf>
- A comparison of hackspace access control systems and membership databases can be found here:
 - <https://docs.google.com/spreadsheets/d/1uXn9LGBIBuJOoRjbfS0Xc9NzUc3StWwt04M8qqtpET8/edit#gid=61365518>
 - and
 - https://wiki.hackerspaces.org/Hackerspace_Software

- Paul has some ideas for a door control system.

Diagrams



Photos



Front panel of the control box, with a button and an override key



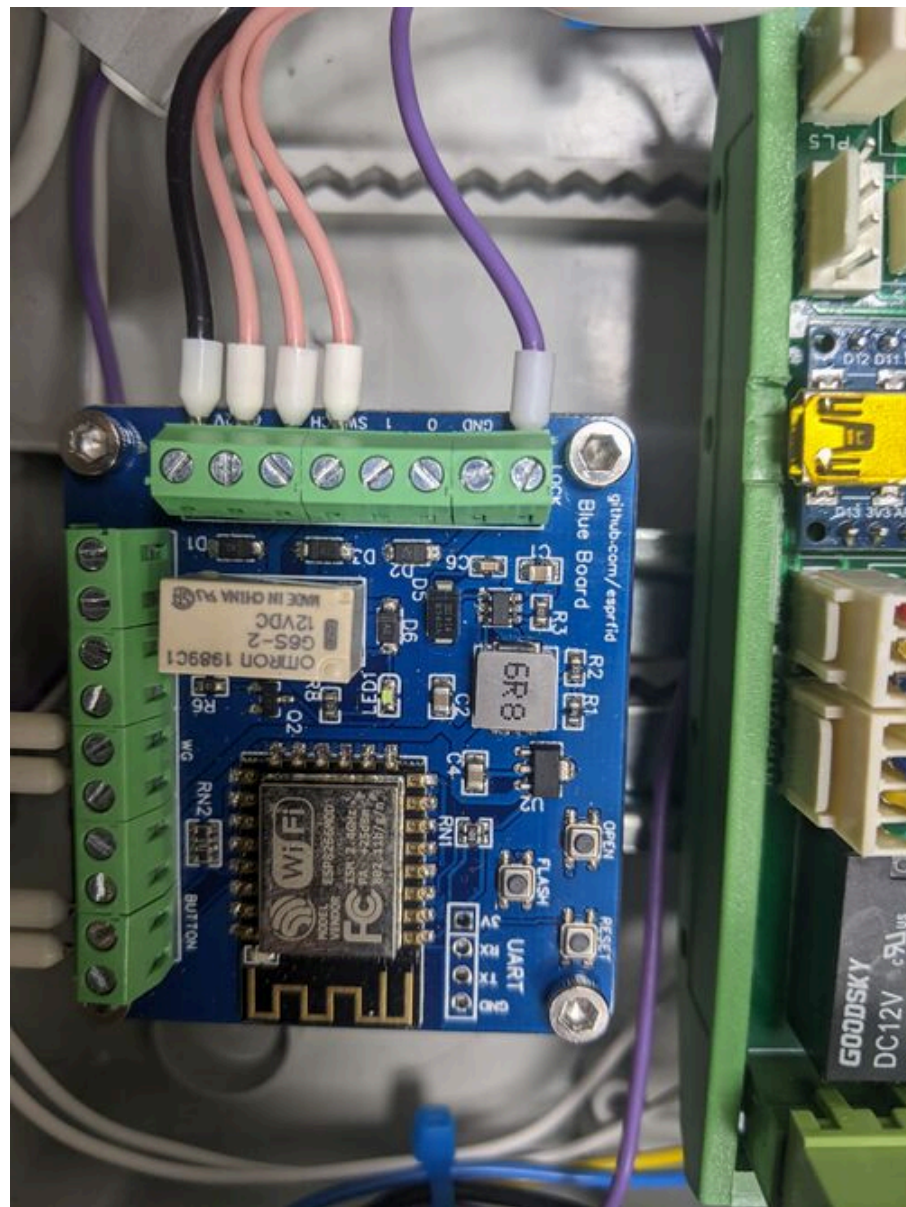
Inside of front panel, showing wiring



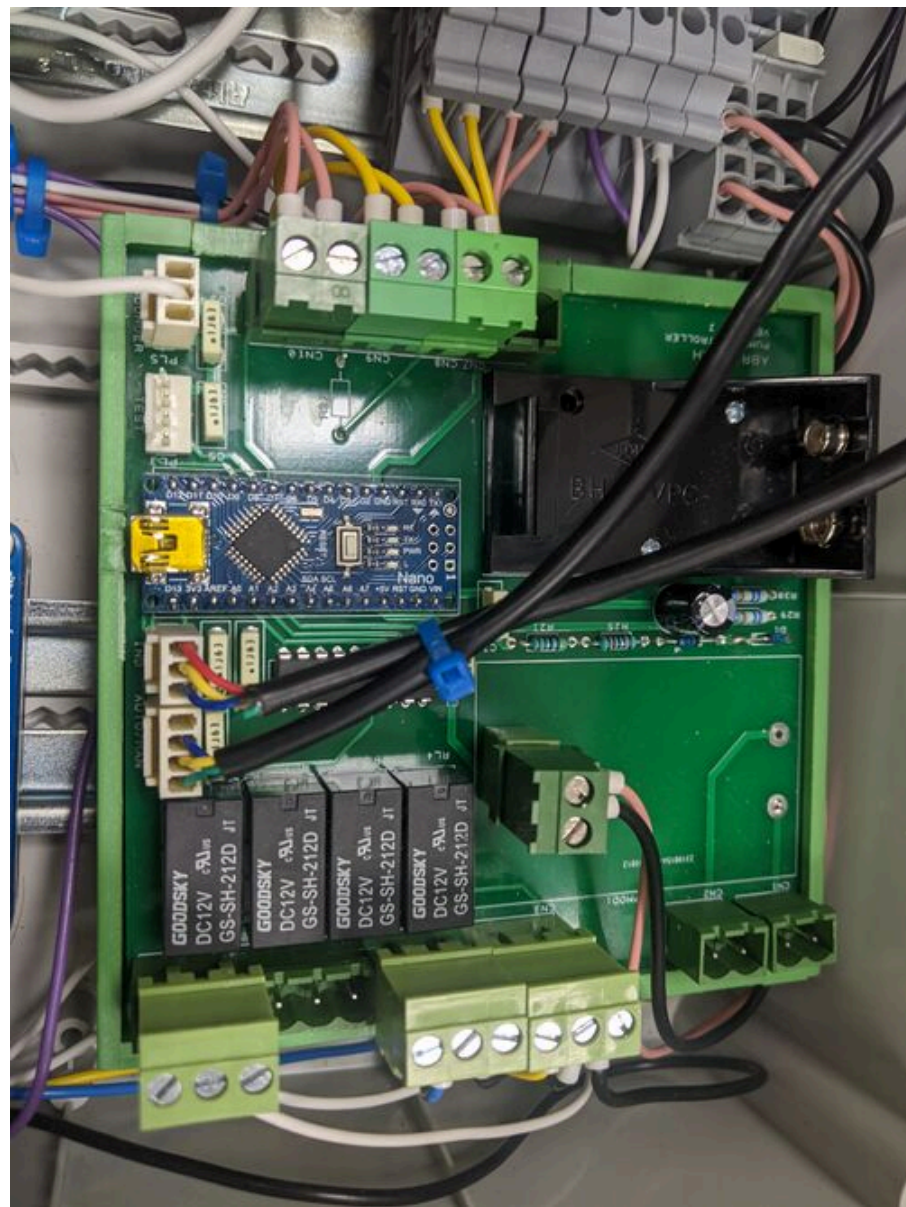
Inside of control panel, with RFID reader at bottom, ESP-RFID control board bottom right, 433MHz interface for lock bottom far right, Arduino door/lock control board top right



Back face of RFID the reader



Close up of ESP RFID access control board



Close up of Arduino door/lock control board



Close up of DESi 433MHz lock interface Module

Door Sensors

Door Closed Sensor

- The door closed sensor is a reed switch activated by a magnet.
- It's connected to a four core cable, with red and yellow wires used for the contacts.

Bolt Lock Sensor

- The lock bolt sensor is a PR12-4DN inductive sensor.
- It needs 10V-30VDC on the brown wire, ground on the blue wire and the black wire is an NPN output.
- Source: <https://www.amazon.co.uk/PR12-4DN-Inductive-Proximity-DC10-30V-Equipment/dp/B08QCT5JQV/>

RFID Reader Module

The RFID Reader connections are:

RFID Connections

Function	Wire Colour
VCC Supply +12V	Red
GND	Black
D0	Green
D1	White
Beep	Yellow
LED	Blue
GND / WG34	Purple

Access Control Database Server System

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This page was last modified on 30 April 2023, at 23:35.